

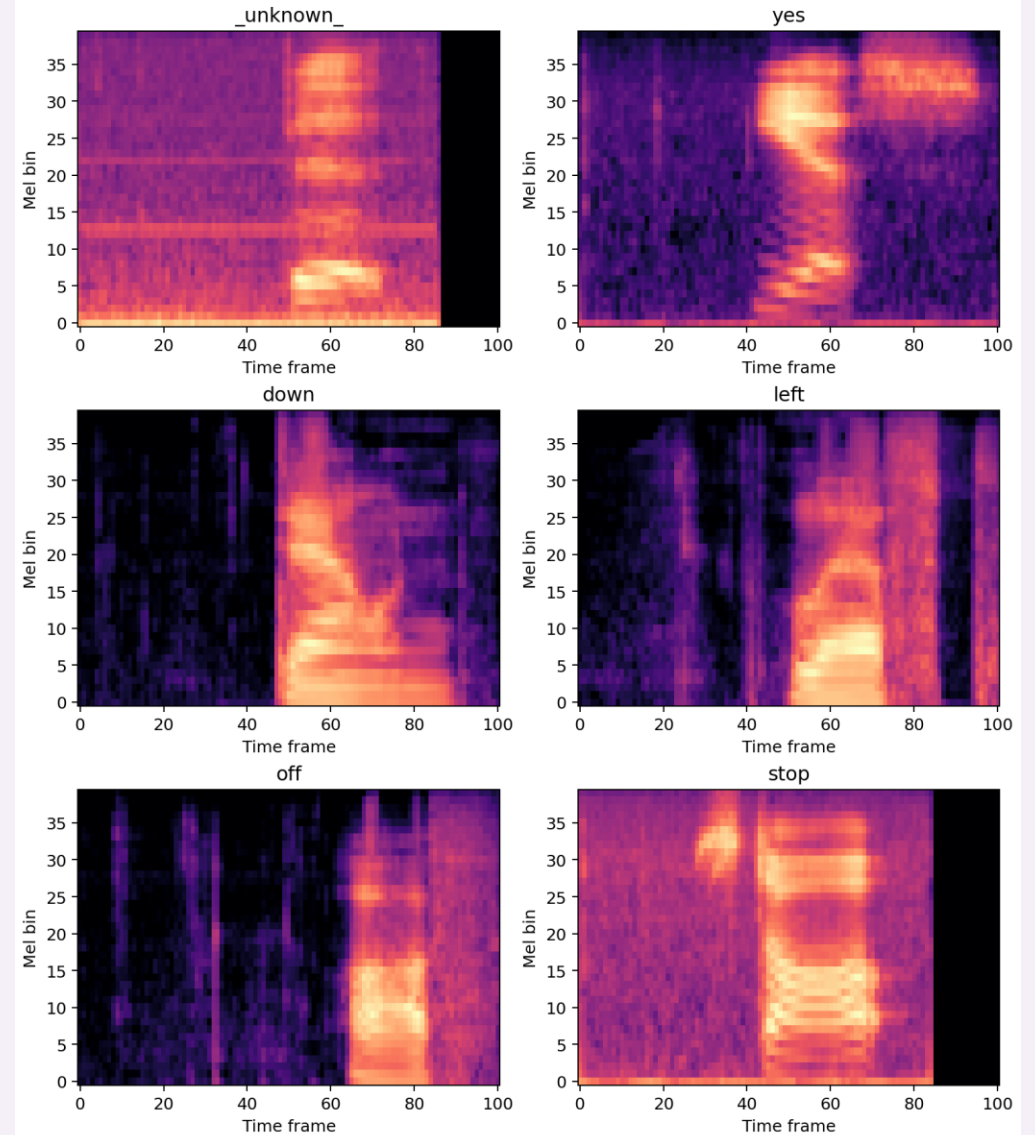
Robust keyword spotting from scratch

A leakage-safe study of audio representations, compact neural networks, and failure modes.

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One training log-Mel spectrogram per modeled class



One second of audio becomes one reliable intent

Input

A mono waveform sampled at 16 kHz

Output

10 commands + unknown + silence

1 s

maximum clip duration

12

balanced modeled classes

0

pretrained components

Educational goal: isolate the impact of representation, architecture, and augmentation without transfer learning.

Balanced modeling preserves the raw audit

64,721

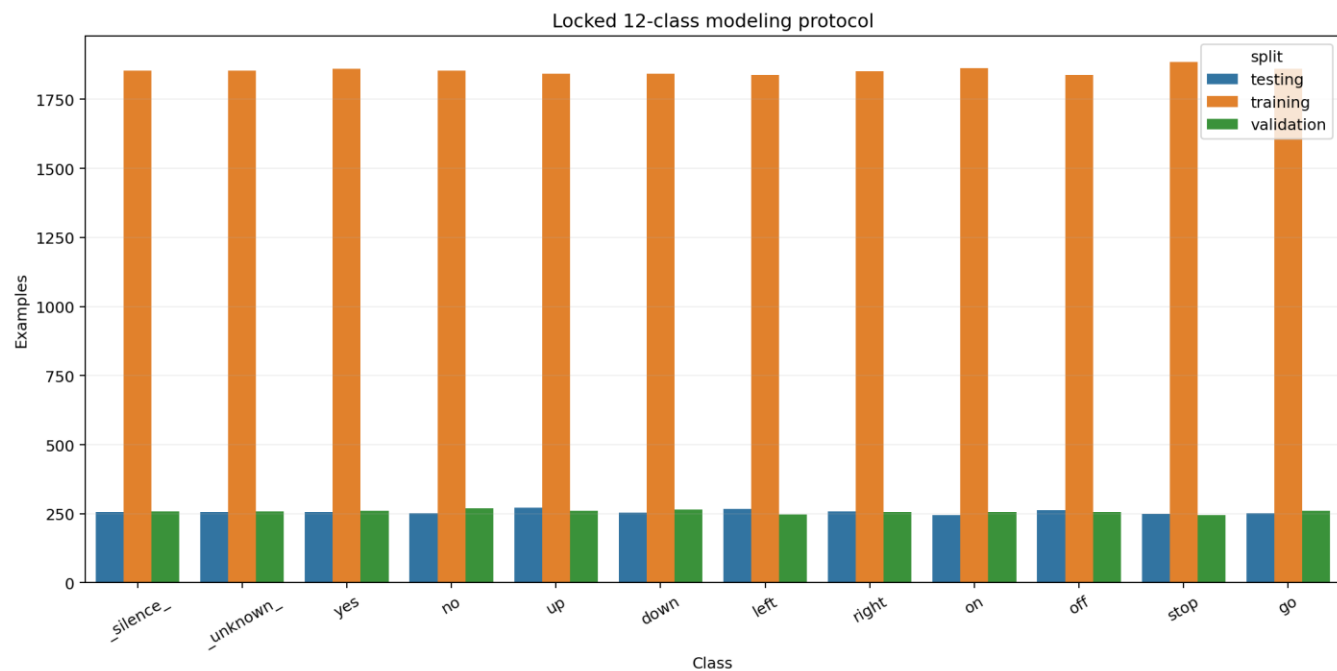
spoken clips in the raw
audit

1,881

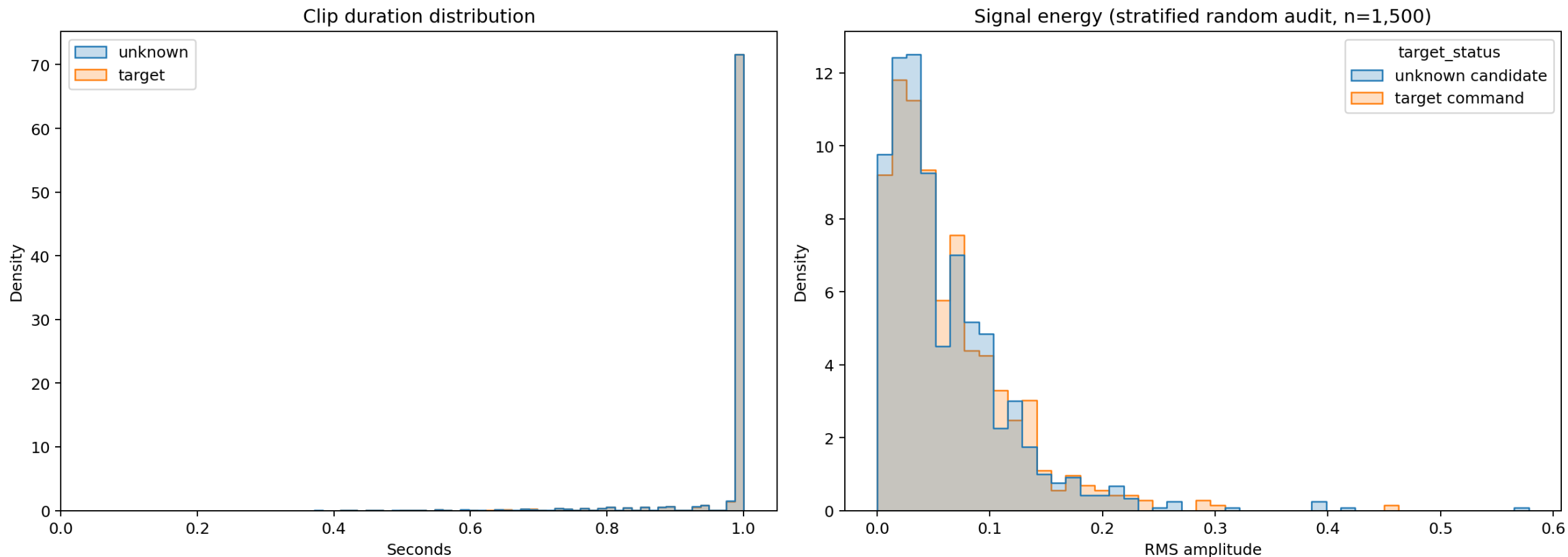
anonymized speakers

28,420 modeled examples

Each class contains the same deterministic unknown and silence proportions across the official splits.



Clip length is stable; signal energy is not



Pad or trim to 16,000 samples; normalize on training data only; augment amplitude, time position, and background noise.

Speaker identity never crosses a split

Official lists define validation and test; all remaining speakers form training. Silence windows are also non-overlapping across splits.

22,246

training examples

3,093

validation examples

3,081

testing examples

Hard guards

- speaker-disjoint validation
- cross-split silence-region checks
- train-only normalization statistics

Fail fast
on any overlap

Log-Mel preserves time-frequency structure



MFCCs are the compact baseline; main experiments retain the full log-Mel map.

Five experiments isolate each design choice

Selection metric: validation macro-F1; the test set remains sealed.

ID	Representation	Architecture	Augmentation	Question
E01	13 MFCC	MLP	None	Can a dense baseline suffice?
E02	40 log-Mel	Small CNN	None	Does local structure help?
E03	40 log-Mel	DS-CNN	None	Can efficiency retain accuracy?
E04	40 log-Mel	DS-CNN	Wave + SpecAugment	What does augmentation trade?
E05	40 log-Mel	Conv + BiGRU	Wave + SpecAugment	Does temporal memory help?

Task structure beats parameter count

MFCC MLP

361,999 params

Flattened coefficients
3 dense blocks
No explicit locality

DS-CNN

46,863 params

Depthwise + pointwise
conv
Global pooling
Compact spatial model

Conv + BiGRU

292,271 params

Convolutional front end
Bidirectional recurrence
Temporal pooling

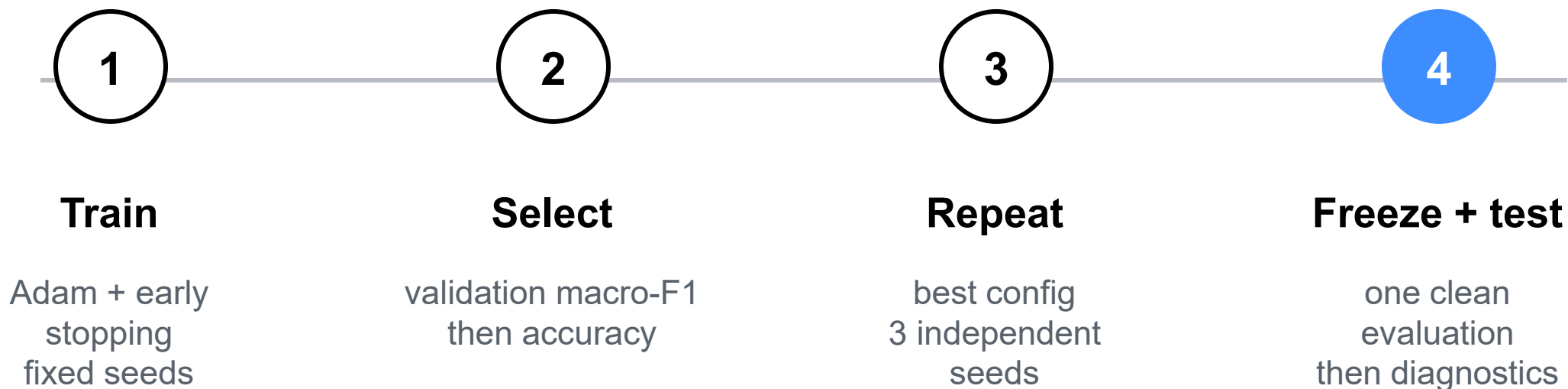
Inductive bias beats raw parameter count.

The test set stays sealed until selection is frozen

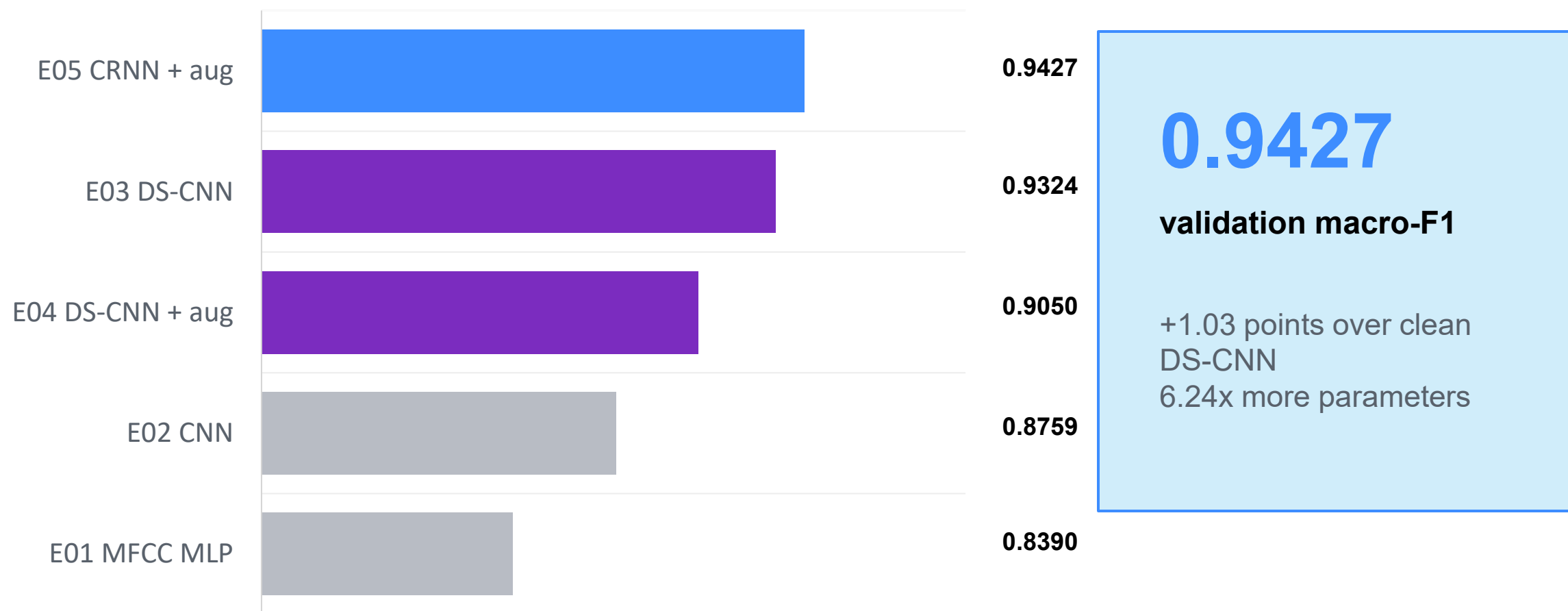
Best-model validation macro-F1 across seeds

0.9424 +/- 0.0009

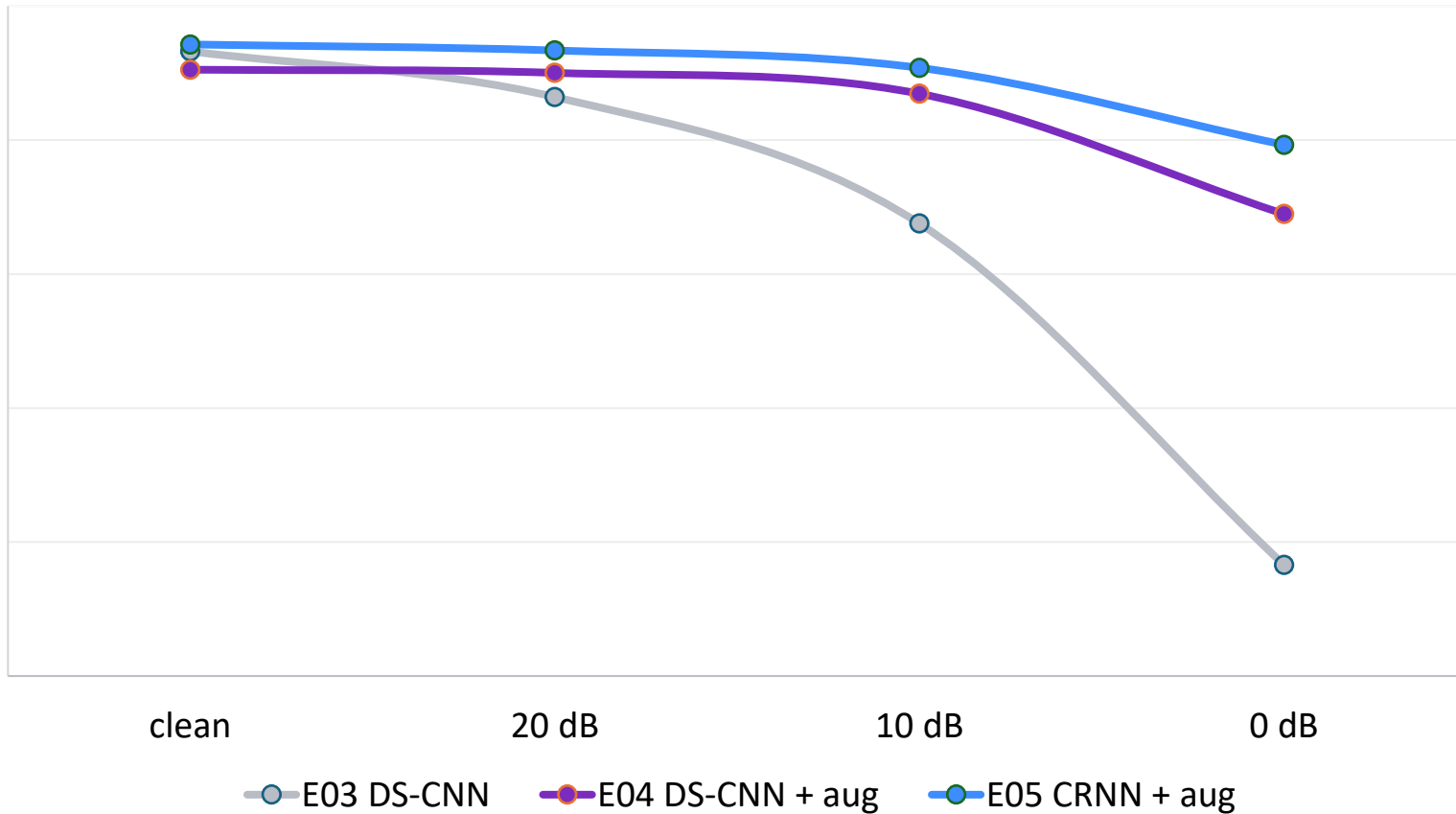
The stability check did not reopen architecture selection.



The CRNN wins validation by one macro-F1 point



Augmentation buys noise robustness



-2.74 pt

clean cost for DS-CNN
augmentation

+52.35 pt

0 dB gain for the same
model

Frozen test confirms the validation result

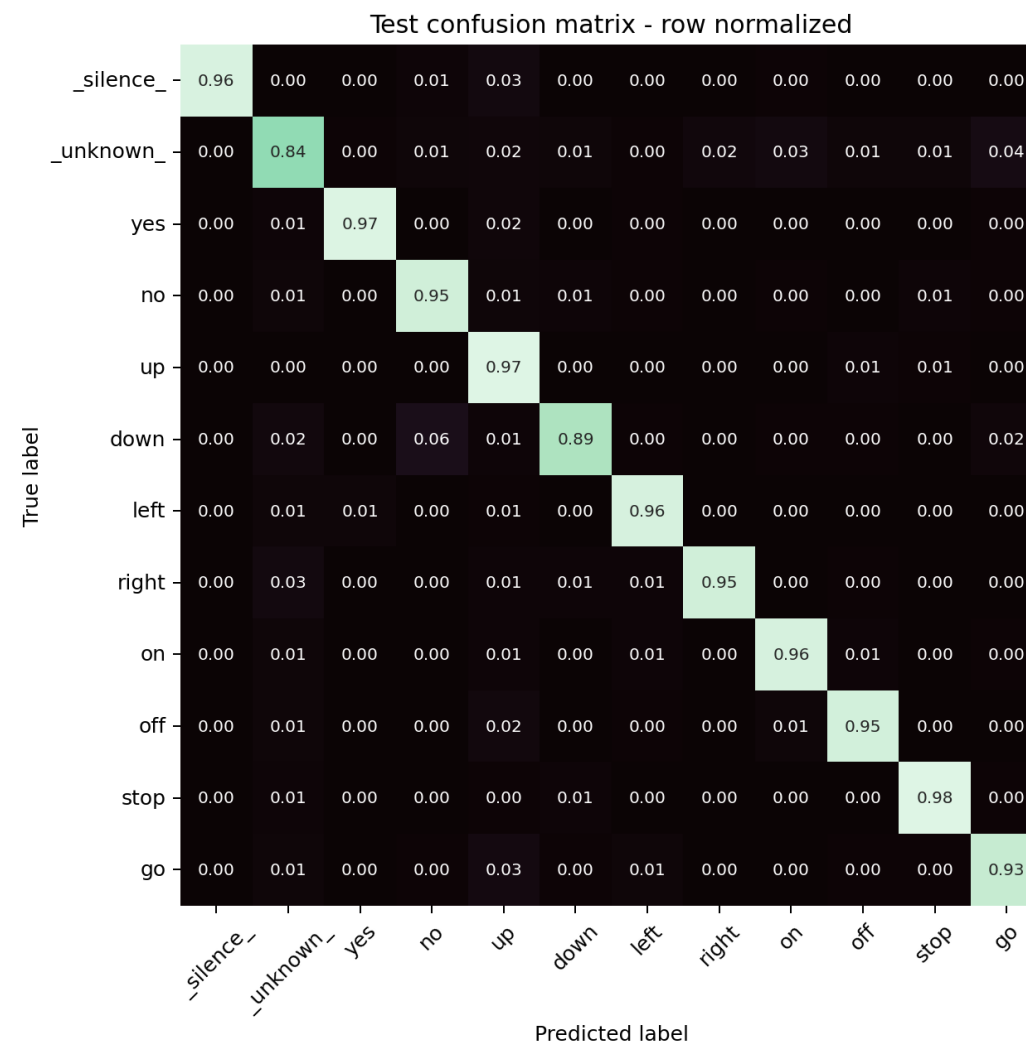
94.19%

test accuracy
95% CI: 93.35-95.00

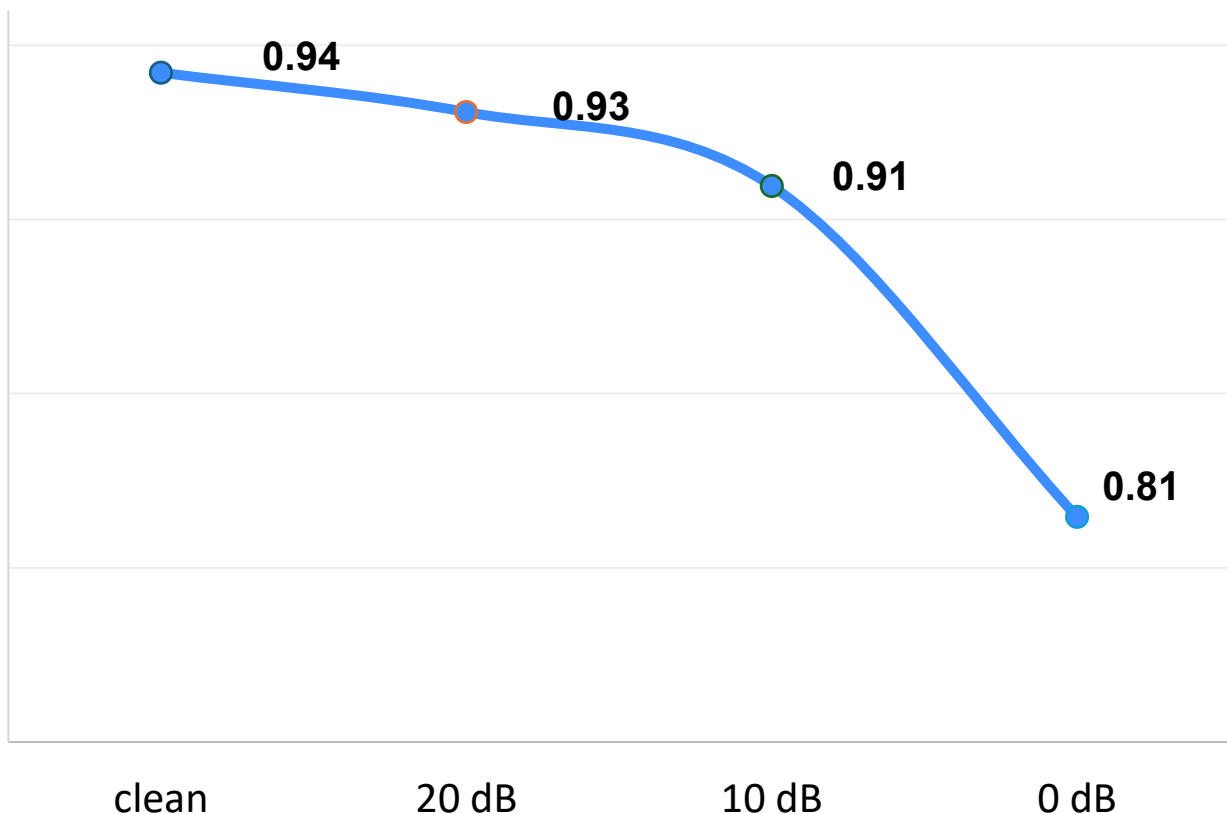
0.9421

test macro-F1
95% CI: 0.9338-0.9501

Validation -> test: -0.06 points



Noise remains the main failure axis



0.0198

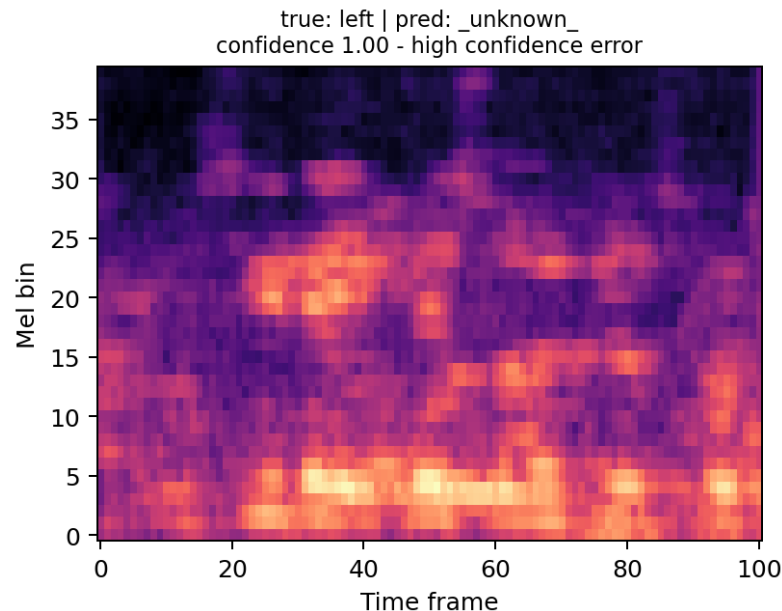
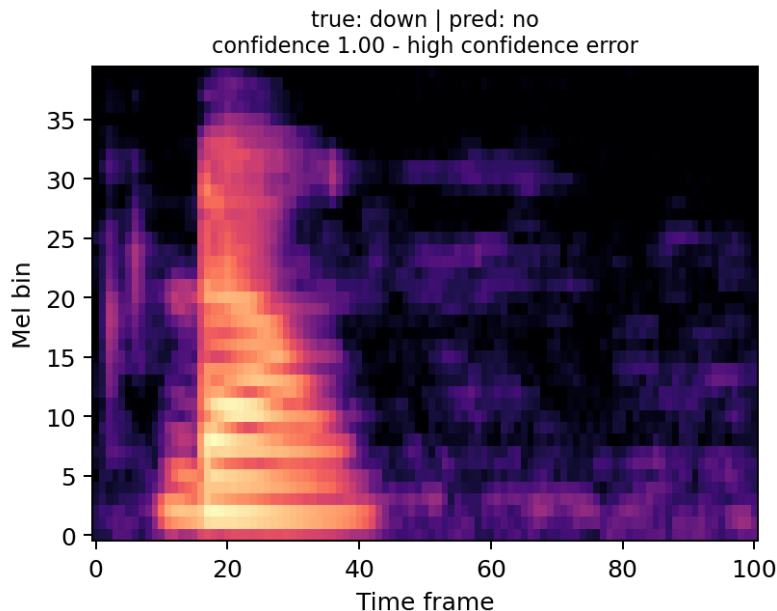
expected calibration error

3.41 ms

median CPU end-to-end latency

**3.42 MiB serialized | +/-100 ms shift costs at
most 0.34 points**

Rare confident failures survive good calibration



Every gallery item links to a playable WAV for listening during the defense.

Weakest class

**unknown F1 =
0.854**

Most common error

**down -> no (14
clips)**

6 confident errors

**all confidence >=
0.998**

Protocol matters as much as architecture

- | | | |
|----|----------------------|---|
| 01 | Structure | Log-Mel convolution plus temporal recurrence reached 0.9421 test macro-F1. |
| 02 | Trade-offs | A 46.9k-parameter DS-CNN remains compelling when footprint matters. |
| 03 | Honest limits | Noise, unknown speech, and rare overconfidence remain the deployment risks. |

Reproducible: manifest, five runs, frozen weights, bootstrap, WAV gallery, CLI demo.